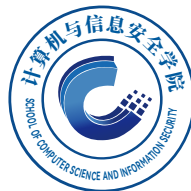


双周工作汇报

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2025 年 10 月 8 日



① 图片总览

② 汇报简介与目标论文背景

③ 问题回答

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Time stamp	Product ID (Code)	Time (s)	Operation	Result	Size (Dimension 1) (mm)	Size (Dimension 2) (mm)
2023/3/1 10:00	0001	152.75	Part	OK	25.0	15.0
2023/3/1 10:00	0002	153.00	Part	OK	25.2	15.1
2023/3/1 10:00	0003	153.25	Part	OK	25.1	15.0
2023/3/1 10:00	0004	153.50	Part	OK	25.3	15.2
2023/3/1 10:00	0005	153.75	Part	OK	25.0	15.1
2023/3/1 10:00	0006	154.00	Part	OK	25.1	15.0
2023/3/1 10:00	0007	154.25	Part	OK	25.2	15.1
2023/3/1 10:00	0008	154.50	Part	OK	25.0	15.0
2023/3/1 10:00	0009	154.75	Part	OK	25.1	15.1
2023/3/1 10:00	0010	155.00	Part	OK	25.2	15.2
2023/3/1 10:00	0011	155.25	Part	OK	25.0	15.0
2023/3/1 10:00	0012	155.50	Part	OK	25.1	15.1
2023/3/1 10:00	0013	155.75	Part	OK	25.2	15.2
2023/3/1 10:00	0014	156.00	Part	OK	25.0	15.0
2023/3/1 10:00	0015	156.25	Part	OK	25.1	15.1
2023/3/1 10:00	0016	156.50	Part	OK	25.2	15.2
2023/3/1 10:00	0017	156.75	Part	OK	25.0	15.0
2023/3/1 10:00	0018	157.00	Part	OK	25.1	15.1
2023/3/1 10:00	0019	157.25	Part	OK	25.2	15.2
2023/3/1 10:00	0020	157.50	Part	OK	25.0	15.0
2023/3/1 10:00	0021	157.75	Part	OK	25.1	15.1
2023/3/1 10:00	0022	158.00	Part	OK	25.2	15.2
2023/3/1 10:00	0023	158.25	Part	OK	25.0	15.0
2023/3/1 10:00	0024	158.50	Part	OK	25.1	15.1
2023/3/1 10:00	0025	158.75	Part	OK	25.2	15.2
2023/3/1 10:00	0026	159.00	Part	OK	25.0	15.0
2023/3/1 10:00	0027	159.25	Part	OK	25.1	15.1
2023/3/1 10:00	0028	159.50	Part	OK	25.2	15.2
2023/3/1 10:00	0029	159.75	Part	OK	25.0	15.0
2023/3/1 10:00	0030	160.00	Part	OK	25.1	15.1

The automated assembly process of the constant velocity joint is shown in <image>, and the production records are provided in <csv file>. Could you help analyze the production cycle time and quality control situation, and also extract the key parameters?



The situation on-site shows that the production cycle time is normal, but there are quality issues with the parts. In particular, Dimension 1 has sudden variations, while Dimension 2 still follows a Gaussian distribution. This suggests that the problem may be concentrated in the process or equipment affecting Dimension 1. Could you please check the relevant processes?

According to the investigation, Dimension 1 represents the pitch circle diameter of the star-shaped raceway. Could you please analyze this based on the statistical data and provide some optimization suggestions?



Since Dimension 1 (the pitch circle diameter) shows a sudden variation while Dimension 2 still follows a Gaussian distribution, it indicates that the issue with Dimension 1 is likely caused by a sudden anomaly in a single process or piece of equipment, rather than a systemic problem. I suggest comparing the raw material batches of the defective parts to check for any batch differences, and also inspecting the wear conditions of molds and fixtures.

图 1: 用于智能制造的多模式大型语言模型系统的对话示例

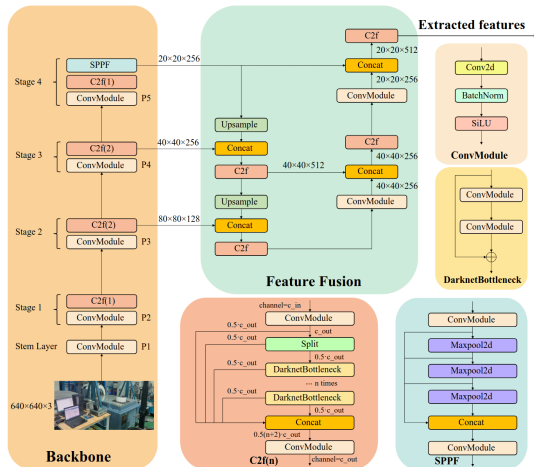


图 2: 基于 CNN 的视觉编码器体系结构的全面说明

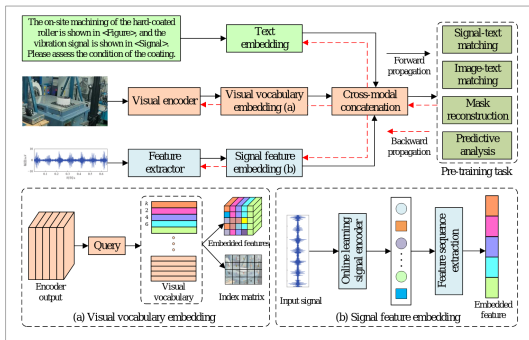


图 3: 用生成动态视觉和信号字典以弥合表征差距的工作流程。

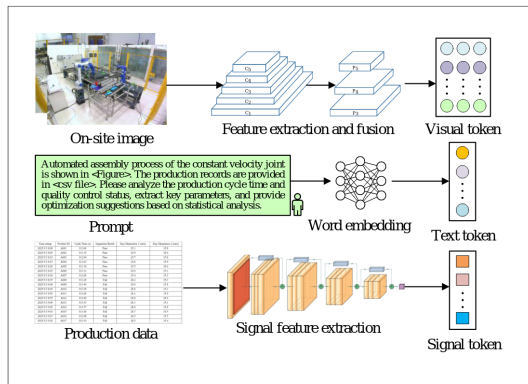


图 4: 调整多种输入方式以输入到基于 Transformer 的语言模型

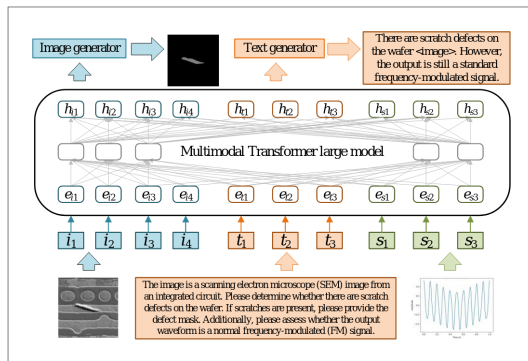


图 5: 基于多模态 Transformer 架构的统一双向图像文本生成框架

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Introduction

经过第二次汇报，在结构优化的前提下，对文章内容的解读仍有缺陷。本次着重讨论如下问题：

- 文章聚焦的核心工程问题是什么？
- 所构建的多模态大模型，其核心决策任务是什么？输出什么？

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Q1

本文要解决的核心工程问题是：基于多源数据（图像、信号、文本等），为工业场景中的设备维护、故障诊断、缺陷检测等任务，提供智能且高效的决策支持方案。如图1所示

Q2

模型需决策的是：对输入的图像、信号、文本等工业数据进行分析与理解，判断设备状态、故障类型或缺陷情况；输出为包含分析结果、诊断结论甚至优化建议的文本或可视化内容（如标注后的图片）。系统整体框架如图5所示

WELLCOME FEEDBACK!

